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A

Magnetic permeability (μ):

~~The magnetic flux intensity over a cross section of unit North pole.~~

The magnetic Induction intensity over in magnetic field

$$\mu = \frac{B}{H}$$

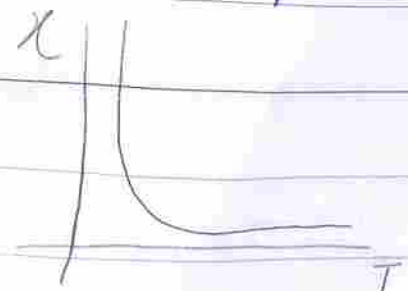
Magnetic susceptibility (χ):

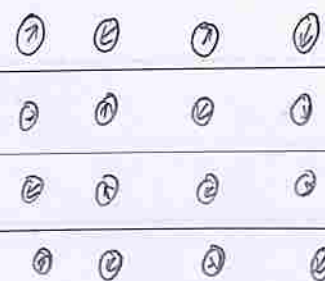
The magnetic Intensity over a cross section of unit North pole

B Paramagnetic materials.

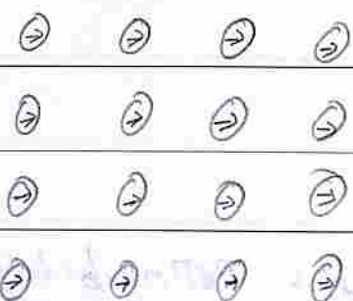
- Dipoles are initially present but all are in random directions.
- When External magnetic force is applied, all the present dipoles slowly turn toward a single direction generating magnetic movement.
- With this generated magnetic movement, the material emits magnetic force. Thus making it a magnetic material.
- The susceptibility is inversely proportional to Temperature

$$\chi \propto 1/T$$





Without external
magnetic force



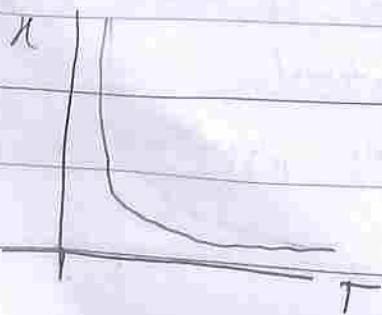
With external
magnetic force

C) Effect of temperature on susceptibility of a ferro and anti ferro magnetic materials.

There are effects of temperature on susceptibility of a ferro and anti ferro magnetic materials.

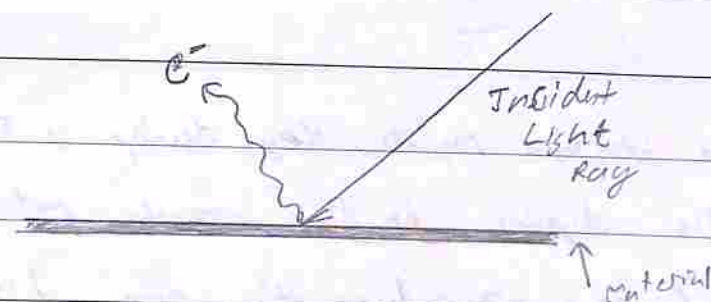
As we know temperature and susceptibility are inversely proportional
 $\chi \propto 1/T$

With the increase of temperature, susceptibility decreases, and with the decrease of temperature, susceptibility increases.



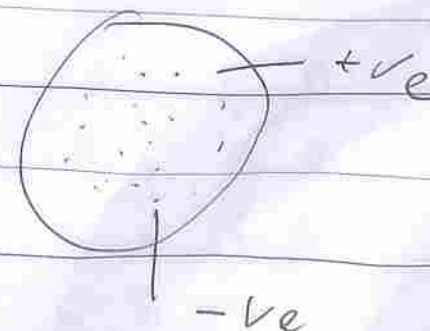
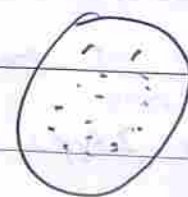
D) Photo electric effect.

The photo electric effect states that when light falls on a photo electric material it emits a matter, that matter is ^{electron} photons. It was first identified by de Broglie.



E) Rutherford atomic model.

Rutherford atomic model was also known as the watermelon model. The whole are is ^{positive} and the ^{negative} e's are scattered in it.



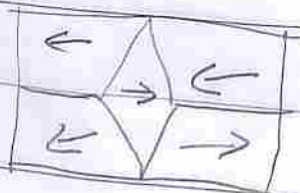
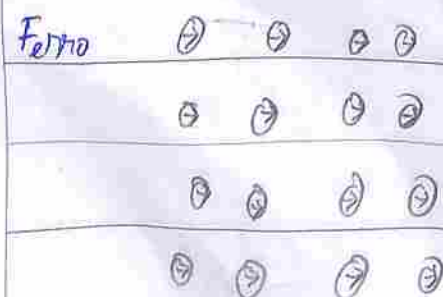
2

A) Importance of Ferri magnetic materials.

Ferri magnetic materials are important to us. They are also known as Spontaneous magnets. These magnets contain dipoles and all are in a single direction. Independent of External magnetic force.

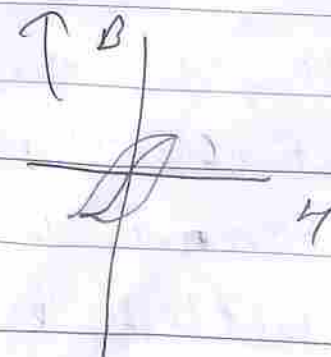
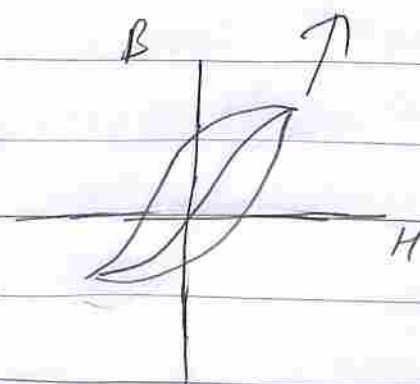
As they are Spontaneous magnets, they develop a concept of Domain theory. The dipoles in the magnetic material form into groups and each group has its own direction. Each group of unidirectional dipoles is called a domain.

They are ^{only} very used with a magnetic like storage devices. They are 2 types in storage, Spontaneous and thin sheet. Thin Spontaneous is like thin needles where each tip of the needle is a dipole. In thin sheets, each grain of the sheet is a dipole.



B) Properties of hard and soft magnetic materials.

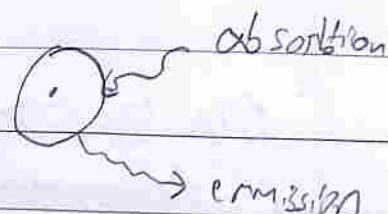
Hard magnetic materials	Soft magnetic materials.
• Large B-H curve volume	• Small B-H curve volume
• Mainly used for permanent magnets	• Mainly used for temporary magnets.
• High Saturation point is large	• Saturation point is short.
• Hard to magnetize and de-magnetize	• Easy to magnetize and de-magnetize



C] Black body radiation.

In classical phy it is said that a material will absorb and ~~then~~ emit at the same time, and so said in theory but failed practically. Quantum physics suggest that emission and absorptions happens at discrete values.

We take a vacuum chamber and place a black body. It is said that this black body can absorb and emit any frequency ~~perfectly~~.



The frequency of absorption, emission and vibration of the body is same..

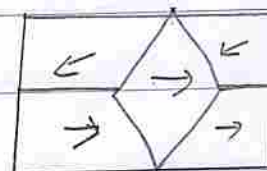
Following these there are 3 laws

- Planck's law
- Stefan's law
- Wien's law
- Rayleigh law.
- Greiner law.

4

A] Domain Theory.

A ferromagnetic material is a spontaneous magnet. thereby it develops a concept of Domain Theory. In the material, the dipoles are divided into groups each having their own directions. These groups of dipoles are called Domains.



We can now introduce an external magnetic force to see further outcome.

Case 1: Weak external magnetic force.

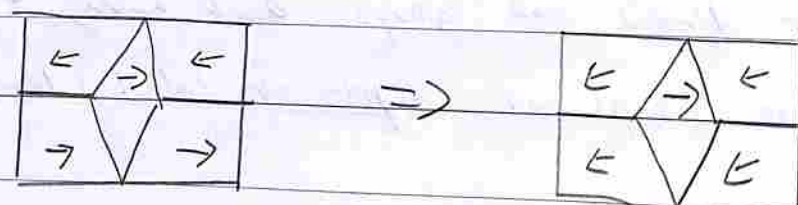
If a weak external magnetic force is introduced to the ferromagnetic material then the dipoles in the domains shift ~~around~~ around changing the size of domains. This effect is reversible.



change in domain size

Case 2: Strong external magnetic force

If a strong external magnetic force is introduced the direction of domain is simply flipped to the side where external magnetic force is pointing to. This is an irreversible action.



change in direction of domain.

B] Different energies involved in growth of domain wall.

- Exchange energy
- Domain wall energy
- Anisotropic energy
- Magnetostriction energy

Exchange energy :- The energies which are due to dipoles

Domain wall energy :- The energy require for external dipole to enter in the domain

Anisotropic energy: The energy for which direction is easy to convert.

easy - (001)

mid - (011)

hard - (111)

Magnetostriction energy :- The total magnetic field after all the changes.

A] De-Broglie's wavelength.

$$K_e :- E = \frac{1}{2}mv^2 \quad E = h\nu \quad \text{--- (1)}$$

$$= \frac{1}{2} \frac{m^2 v^2}{m}$$

$$E = \frac{1}{2m} p^2 \quad \text{--- (2)}$$

$$p = \sqrt{2mE}$$

$$\text{① and ②} \Rightarrow h\nu = \frac{p^2}{2m} = \frac{h^2}{\lambda^2 2m} \Rightarrow \lambda = \frac{h}{\sqrt{2mE}}$$

PO:- $q = eV$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$T: \lambda = \frac{h}{\sqrt{3kT}}$$

1] F] Physical Significance of wave function

The wave function is used to represent the function of the wave. It can be denoted by many various ways.

It's physical significance is that it is the function of the wave in wave function.

